From force diagrams to motion

antocciphysics | TEACHER KEY | CPA

Learning target: Use a force diagram to predict acceleration and velocity change.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 13; 3 course days later.*
Net force 15 N right; acceleration 5 m/s² right.

02. Explain a worked example (worked)
Net force 4 N right; a=4/2. Straight v-t line from 0 to 6 m/s over 3 s.

03. Horizontal case (model)
30-10=20 N right; a=4 m/s² right.

04. Use a given ramp component (practice)
Net 6 N down ramp; a 3 m/s² down ramp.

05. Moving up, accelerating down (reason)
Its speed decreases because acceleration opposes up-ramp velocity.

06. Choose a method (mixed) *Review: Unit 1, Day 4; 12 course days later.*
Displacement -35 m. Second arrow points southeast with component ratio 9: 4.

07. Exit check (exit)
Net 9 N right; a 3 m/s² right; leftward speed decreases.

08. STRETCH: Apply it another way (stretch)
Acceleration halves to 1 m/s²; final speed 3 m/s; graph slope halves.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 13; 3 course days later.*
Net -21 N; a=-7 m/s². Need current velocity: leftward motion speeds up; rightward motion slows.

02. Explain a worked example (worked)
Net force 4 N right; a=4/2. Straight v-t line from 0 to 6 m/s over 3 s.

03. Angled pull (model)
N=49-12=37 N; net x 15 N; a_x 3 m/s² right.

04. Frictionless ramp (practice)
N=mg cos 20=18.4 N; a=g sin 20=3.35 m/s² down.

05. Oppose upward motion (reason)
a=(-4×9.8×.5-5)/4=-6.15 m/s²; slowing while moving up.

06. Choose a method (mixed) *Review: Unit 1, Day 4; 12 course days later.*
Displacement 15 m; distance 55 m. An angle needs a reference direction/axis and rotation sense.

07. Exit check (exit)
Weight parallel 9.8 N; net 6 N down; a 3 m/s² down; N=16.97≈17.0 N.

08. STRETCH: Apply it another way (stretch)
a=-2; turns at 4 s; v_f=-4 m/s; displacement 48-36=12 m; distance 16+4=20 m.